



EchoSync

AI-Assisted Pre-Arrival Intelligence & First Response

A privacy-first, deployment-ready emergency response ecosystem designed specifically for Singapore's high-density HDB urban environments.

The “Silent Emergency” Crisis

Singapore’s emergency response is highly effective **once a call is made**.

But in unwitnessed home emergencies, the fatal gap happens **before the call exists**.

For seniors living alone, there may be:



No bystander



No panic button pressed



No timely 995 call



News Evidence

Real local cases show the risk of emergencies being discovered too late.

"Elderly residents found motionless in HDB flat" — CNA, 2026



The danger is not just the emergency — it is when no alert is raised.



Re-Engineering the Response Model



The Old: Reactive Reporting



Resident must press button or call



Neighbour discovery may be delayed



SCDF receives limited pre-arrival context



Manpower wasted by false alarms



The New: Preventive Intelligence



AI detects correlated emergency signals



Confidence score reduces false alerts



Dispatcher verifies before escalation



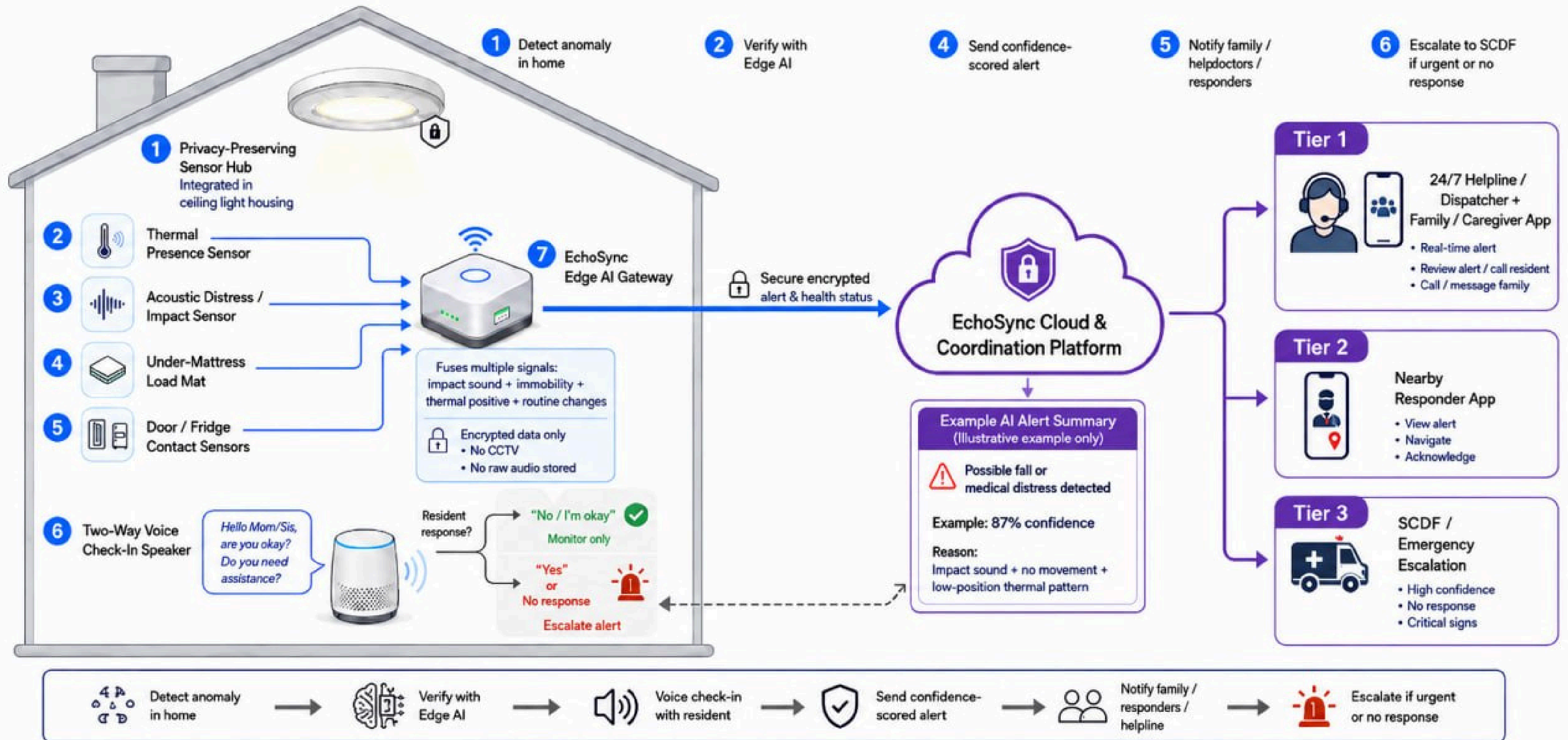
CFR + AED support starts earlier



From reactive response to proactive prevention of late discovery.

EchoSync System Workflow

AI-assisted fall verification, voice check-in, and responder coordination for seniors living alone



Accelerated Intervention Timeline



▶▶ EchoSync shortens the silent window before first human action.

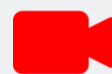
Dispatcher-Verified CFR Routing

✗ Current Limitations



Buttons & Wearables

Fail if the resident is unconscious. Require constant charging and conscious physical action to activate.



Surveillance Cameras

Severe privacy violations. Unacceptable for large-scale deployment inside private HDB bedrooms and living spaces.



myResponder App

myResponder is highly effective, but still needs an emergency alert first.



The EchoSync Advantage



Passive Ambient Sensing

Detects falls and immobility automatically. Protects residents even when they are unconscious and alone.



Privacy-first / PDPA-conscious design

Uses non-visual thermal and acoustic edge AI. No raw video or private conversations are ever stored.



Automated CFR Routing

The AI creates the first credible alert, bridging the gap and triggering myResponder before 995 is called.



EchoSync is not another button — it is the missing AI first-mile layer.



Built for Singapore's HDB Ecosystem



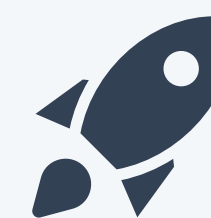
Existing Infrastructure

Leverages standard HDB layouts, the active AED network, and Smart Nation readiness.



Minimal Disruption

Ceiling-mounted nodes requiring no major renovations. Fully PDPA-compliant deployment.



Pilot-Ready Rollout

Targeted for HDB Silver Zones with opt-in households. Community-first response under SCDF supervision.

 Designed to augment — not replace — existing SCDF workflows

🎯 Operational Impact & Analytics

↑ Increase



Detection-to-Alert Delay

Closes the unwitnessed gap from hours to <60s.



Earlier CFR CPR Intervention

Community chest compressions begin before SCDF arrival.



Faster AED Coordination

myResponder routes volunteers to nearest corridor AED.

↓ Decrease



SCDF Situational Awareness

Paramedics arrive knowing exact anomaly context.



Dispatcher Uncertainty

Explainable AI reveals exactly why alerts triggered.



False Dispatch Burden

Multi-modal AI & grace period filter accidental triggers.



Every verified alert becomes learning data to improve thresholds, response planning, and future deployment.



Thank You